

Final Report

Abstract

Diabetes remains a major public health concern in the United States, making early identification of at-risk individuals a key priority. This project uses the 2015 BRFSS Diabetes Health Indicators dataset to examine behavioral, demographic, and clinical predictors of diabetes status and to compare two modeling approaches: multinomial logistic regression and binary logistic regression. Exploratory data analysis confirmed meaningful variability across predictors and revealed class imbalance, particularly for the prediabetes category. The multinomial model, after up-sampling to address this imbalance, offered more detailed insight by distinguishing among No Diabetes, Prediabetes, and Diabetes, though its overall predictive performance was moderate. The binary logistic regression model produced stronger overall performance, achieving higher accuracy, better model fit, and a strong AUC, while maintaining interpretability. Key predictors across both models—including BMI, age, high blood pressure, high cholesterol, general health, and physical activity—aligned closely with pre-existing literature on diabetes risk factors. Although limited by self-reported data, class imbalance, and the exclusive use of regression-based models, this analysis demonstrates consistent patterns in diabetes prediction and highlights the trade-off between model detail and predictive strength.

Introduction

Diabetes is one of the most significant and chronic health conditions in the United States, affecting more than 38 million adults (Centers for Disease Control and Prevention [CDC], 2024) and contributing to long-term health and economic burdens. Identifying individuals who are at risk—particularly those in the transitional state of prediabetes—is a huge priority because early

detection and intervention can dramatically reduce progression to Type 2 diabetes. Large population-based surveys, like the one covered in the analysis, provide an opportunity to analyze behavioral, demographic, and clinical factors that contribute to diabetes risk and to develop predictive models that can support diabetes prevention efforts.

This project uses the 2015 Behavioral Risk Factor Surveillance System (BRFSS) “Diabetes Health Indicators” dataset (Teboul, 2019), which is a nationally representative survey containing nearly 250,000 observations on health behaviors, chronic conditions, and demographic characteristics. The dataset includes a three-level diabetes classification (no diabetes, prediabetes, diabetes) along with relevant predictors such as BMI, age, physical activity, smoking status, blood pressure, cholesterol, general health ratings, and more.

The goal of this project is to model diabetes status using two regression approaches: multinomial logistic regression, which focuses on three states of diabetes (no risk, prediabetes, diabetes), and binary logistic regression, which combines prediabetes and diabetes into a single at-risk category. The motivation for using two models is to compare whether the added complexity of a multinomial framework provides more meaningful insights or whether a simpler binary approach offers better predictive performance. This project aims to better understand the drivers of diabetes risk and to assess which modeling strategy is most effective for this data.

Literature Review

There has been a lot of research done already on diabetes and what risk factors may cause it. In this literature review, three studies will be covered and connected to the scope of this project. In 2015, Okwechime et al. (Okwechime, Roberson, & Odoi, 2015) conducted a large BRFSS-based analysis using multinomial logistic regression to model three outcomes: no

diabetes, prediabetes, and diabetes. Their results showed that higher BMI, older age, hypertension, high cholesterol, and worse self-reported general health significantly increased the relative risk of both prediabetes and diabetes compared to no diabetes. Importantly, their study demonstrated that prediabetes and diabetes do not share identical predictors. Some factors had stronger effects for diabetes than for prediabetes, which highlights the value of keeping the three categories separate rather than combining them. This study utilized the multinomial logistic regression model and confirmed the model's capabilities of fitting a BRFSS dataset.

In 2010, Yang et al. (Yang et al., 2010) used BRFSS data to examine how cardiovascular health indicators relate to the prevalence of prediabetes and diabetes. They found that obesity, older age, smoking, physical inactivity, and lower socioeconomic status were strongly associated with diabetes, reinforcing the inclusion of these behavioral and clinical variables in this project.

In 2022, Ullah et al. (Ullah et al., 2022) extended BRFSS-based diabetes modeling by comparing logistic regression to modern machine-learning methods. They found that although machine-learning models sometimes achieve higher accuracy, traditional logistic regression performed very well and was much easier to interpret. Their results also confirmed that BMI, hypertension, cholesterol, physical activity, and general health consistently show the strongest associations with diabetes, regardless of model type.

These three studies collectively justify both the predictors and modeling strategies used in this project. They confirm that BRFSS health-indicator variables are strong predictors of diabetes and that logistic regression models are appropriate for analyzing them. They also highlight the value of comparing multinomial and binary logistic models, as the multinomial model offers more detailed insight while the binary model provides stronger predictive performance. This study contributes to this literature by applying both modeling approaches to the same BRFSS dataset

and evaluating them side-by-side using accuracy, pseudo R-squared, and AUC. This direct comparison aims to clarify whether the additional complexity of modeling prediabetes as a standalone class leads to more accurate predictions (multinomial modeling), or whether a simpler binary classification is sufficient for this data.

Exploratory Data Analysis

The Diabetes Health Indicators (BRFSS 2015) dataset was first explored to understand independent variable structure, response variable distribution, and general patterns before model building. The dataset contains 253,680 complete observations with no missing values, so no cleaning or imputation was required. The outcome variable of interest, diabetes, consists of three categories: No Diabetes, Prediabetes, and Diabetes.

To explore the continuous and ordinal predictors, descriptive statistics were generated for BMI, Age, General Health, Physical Health Days, Mental Health Days, Education, and Income. These summaries showed wide variability in BMI (figure 1), a distribution of age categories skewed toward older adults, and general self-reported health ratings clustered around good/very good. Physical and mental health days were heavily right-skewed, with many respondents reporting zero days of poor health.

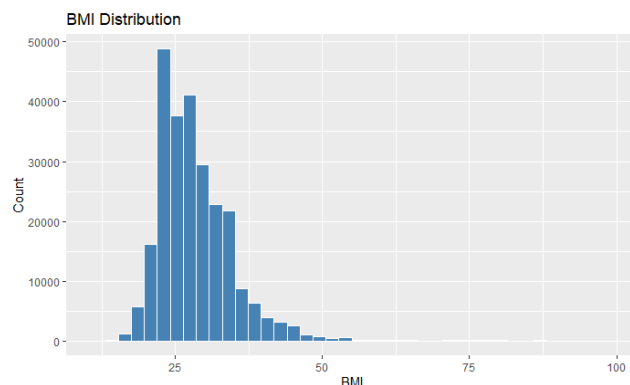


Figure 1: Distribution of BMI values showing wide variation across respondents.

Binary predictors (examples: HighBP, HighChol, PhysActivity, Smoker, DiffWalk) were examined using frequency tables and bar plots. This was to see how each binary variable was distributed. The bar plot for HighBP (figure 2) is below:

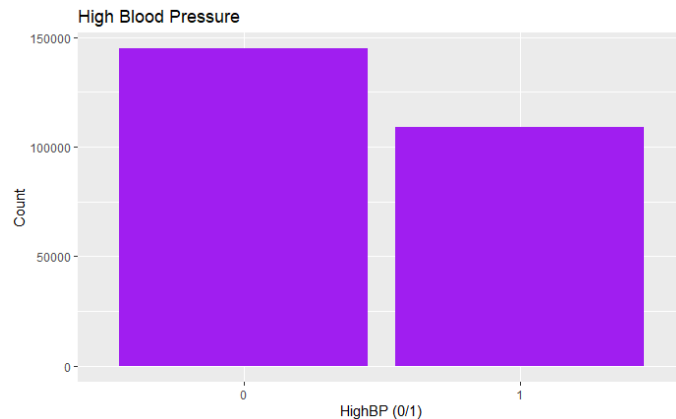


Figure 2: High blood pressure counts illustrating imbalance across binary categories.

A correlation heatmap (figure 3) was constructed for the numeric predictors to evaluate potential collinearity before regression modeling. All correlations remained below approximately 0.60, leading to no indication of multicollinearity and supporting the inclusion of all predictors in both regression models.

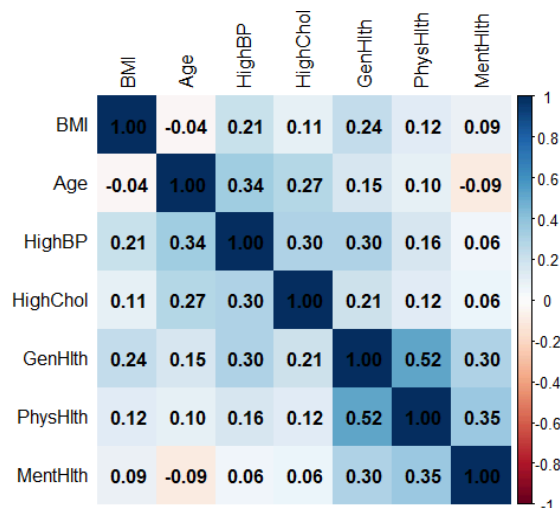


Figure 3: Correlation heatmap showing no strong multicollinearity among predictors.

This exploratory analysis provided evidence that all of the predictors exhibit meaningful variability/significance and are suitable for logistic and multinomial modeling. Backward or piecewise variable selection will not be necessary.

Results

Two models were fit to predict diabetes status: a multinomial logistic regression splitting diabetes among three categories (none, pre, and diabetes), and a binary logistic regression combining prediabetes and diabetes into one “at-risk” class. Each model’s performance, strengths, and limitations are covered in this section.

Multinomial Logistic Regression

The initial multinomial model struggled to identify the prediabetes class due to severe class imbalance. After up-sampling the training data, performance became more balanced across classes. The model achieved moderate detection of Prediabetes and Diabetes and produced an interpretable confusion matrix (figure 4).

```
Confusion Matrix and Statistics

          Reference
Prediction NoDiabetes Prediabetes Diabetes
NoDiabetes  42570      385      1911
Prediabetes 10490      400      2376
Diabetes    11050      604      6316

Overall Statistics

          Accuracy : 0.6476
          95% CI : (0.6442, 0.651)
        No Information Rate : 0.8424
        P-Value [Acc > NIR] : 1

          Kappa : 0.2459

McNemar's Test P-Value : <2e-16

Statistics by Class:

          Class: NoDiabetes Class: Prediabetes Class: Diabetes
Sensitivity          0.6640          0.287977          0.59568
Specificity          0.8085          0.827794          0.82207
Pos Pred Value       0.9488          0.030152          0.35147
Neg Pred Value       0.3104          0.984261          0.92625
Prevalence           0.8424          0.018252          0.13933
Detection Rate       0.5594          0.005256          0.08299
Detection Prevalence 0.5896          0.174319          0.23613
Balanced Accuracy     0.7363          0.557886          0.70888
```

Figure 4: Confusion matrix for balanced multinomial model classification.

Odds ratios (figure 5) indicated that BMI, age, high blood pressure, high cholesterol, difficulty walking, and worsening general health increased the odds of both prediabetes and diabetes relative to no diabetes, while physical activity, education, and income reduced risk. These directions were consistent with expectations from accepted diabetes epidemiology literature.

	(Intercept)	BMI	Age	PhysActivity	Smoker	HighBP	HighChol
Prediabetes	0.021912115	1.068307	1.158893	0.9760172	0.9423041	1.461647	1.784628
Diabetes	0.002530685	1.086689	1.192098	0.9316709	1.0215686	2.163149	1.946302
	GenHlth	PhysHlth	MentHlth	Diffwalk	Income	Education	
Prediabetes	1.443966	0.9942954	1.0081826	0.9193728	0.9238486	0.9394446	
Diabetes	1.874922	0.9942922	0.9956616	1.0667899	0.9340952	0.9706618	

Figure 5: Odds ratios from multinomial model highlighting major predictor effects.

Model fit was evaluated using McFadden's pseudo R-squared, yielding a value of 0.128, consistent with moderate explanatory power for large survey-based datasets. The model accuracy concluded in the confusion matrix was 0.6476, which is also moderate explanatory power.

Binary Logistic Regression

The binary logistic regression model produced stronger overall predictive performance. Its accuracy was higher than the multinomial model, and the model achieved a strong AUC of 0.8125, demonstrating effective classification between individuals with and without diabetes/prediabetes. The ROC curve for the binary logistic regression model is below in figure 6.

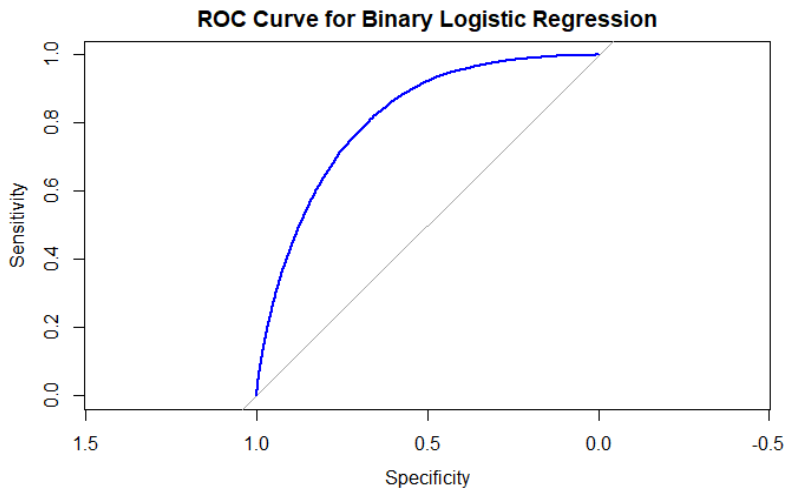


Figure 6: ROC curve for binary logistic regression demonstrating strong AUC performance.

The confusion matrix (figure 7) reflected high sensitivity for identifying individuals without diabetes and lower specificity for the at-risk group due to class imbalance. The accuracy (0.8469) was higher than the multinomial regression model (0.6476).

Confusion Matrix and Statistics

```
Prediction      Reference
              0      1
0      62375   9915
1      1735   2078

      Accuracy : 0.8469
      95% CI   : (0.8443, 0.8495)
      No Information Rate : 0.8424
      P-value [Acc > NIR] : 0.0003134

      Kappa : 0.2023

      Mcnemar's Test P-value : < 2.2e-16

      Sensitivity : 0.9729
      Specificity : 0.1733
      Pos Pred Value : 0.8628
      Neg Pred Value : 0.5450
      Prevalence : 0.8424
      Detection Rate : 0.8196
      Detection Prevalence : 0.9499
      Balanced Accuracy : 0.5731

      'Positive' Class : 0
```

Figure 7: Binary model confusion matrix showing accuracy and class-specific performance.

Jack Griffin
STA4173

Pseudo R^2 for the binary model was 0.199, indicating stronger explanatory power than the multinomial model. The effects of predictors remained consistent across models, reinforcing those directional relationships.

Conclusion

Based on the overall performance and interpretability of both models, the binary logistic regression model would be the preferred choice for predicting diabetes risk in this dataset.

Although the multinomial model provided more detailed information by separating prediabetes and diabetes, the binary model demonstrated stronger predictive performance, including higher overall accuracy, substantially better model fit (pseudo R^2), and a strong AUC, while still identifying the same key behavioral and clinical risk factors highlighted in the related literature.

These findings support prior public-health research showing that BMI, age, cardiovascular indicators, and general health are reliable predictors of diabetes in BRFSS data. Several limitations should be stated: the dataset is self-reported, the outcome distribution is highly imbalanced, and only regression-based models were assessed. Future improvements could include exploring alternative resampling methods, adding interaction terms, or comparing logistic regression to other machine learning models.

Jack Griffin

STA4173

Works Cited

Centers for Disease Control and Prevention. (2024). *National diabetes statistics report, 2024*.

U.S. Department of Health and Human Services.

<https://www.cdc.gov/diabetes/php/data-research/index.html>

Okwechime, I. O., Roberson, S., & Odoi, A. (2015). Prevalence and predictors of pre-diabetes and diabetes among adults in Florida: A multinomial logistic modeling approach. *PLOS ONE*, 10(12), e0145781. <https://doi.org/10.1371/journal.pone.0145781>

Teboul, A. (2019). *Diabetes health indicators dataset* [Data set]. Kaggle.

<https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>

Ullah, I., Ashraf, I., Mehmood, A., Choi, G. S., & Baik, S. W. (2022). A hybrid prediction model for diabetes using machine learning methods. *Healthcare*, 10(1), 36.

<https://pubmed.ncbi.nlm.nih.gov/36210985/>

Yang, Q., Yu, D., Zabetian, A., Gregg, E. W., Albright, A., & Imperatore, G. (2010). Ideal cardiovascular health and the prevalence of prediabetes and diabetes among U.S. adults.

Preventing Chronic Disease, 7(2), A47. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2894805/>

We, the project teams members, certify that below is an accurate account of the percentage of effort contributed by each team member in the project and report.

<u>Project Team Member</u>	<u>Percentage of Total Effort</u>
Jack Griffin	100%